Cogtics – Phase-by-Phase Backend Development Plan

Detailed Engineering Execution Roadmap

Phase 0 – Foundations

Set up monorepo, CI/CD, environment configuration, secrets management, database migrations, tenant isolation model, RBAC, audit logging primitives, job queue, and object storage.

Phase 1 – Connector Framework

Implement standardized connector interfaces with OAuth, schema discovery, incremental sync, rate-limit handling, retries, logging, and health checks. Build Stripe and Salesforce first.

Phase 2 – Canonical Data Model & Ingestion

Design and implement canonical tables. Build ingestion pipeline stages: fetch, validate, normalize, identity resolution, idempotent upsert, and snapshot creation.

Phase 3 – Identity Resolution

Implement canonical customer model, deterministic and assisted matching rules, human confirmation queue, and suppression logic for unresolved matches.

Phase 4 – Remaining Integrations

Add HubSpot, QuickBooks, and DocuSign connectors. Implement CSV and PDF manual upload ingestion mapped into the same canonical pipeline.

Phase 5 – Contract Intelligence

Implement contract storage, versioning, clause-level term extraction, confidence scoring, override workflows, and contract readiness gating.

Phase 6 – Revenue Recognition Engine

Build performance obligation model, transaction price allocation, straight-line recognition schedules, deferred vs recognized tracking, and reconciliation outputs.

Phase 7 – Detection Engine

Implement deterministic revenue leakage and rev-rec mismatch rules. Build issue generation, severity scoring, confidence calculation, and workflow state transitions.

Phase 8 – Evidence Generation

Generate immutable evidence packs including calculations, clause references, source links, and completeness scoring tied to snapshots.

Phase 9 – Simulation Engine

Implement snapshot cloning, override application, delta computation, multi-fix simulations, template saving, and result persistence.

Phase 10 – Exports & Digests

Build CSV exports, server-rendered PDF generation, scheduled email digests, and export audit logs.

Phase 11 – Observability & Hardening

Implement monitoring, alerts, retries, dead-letter queues, backfill testing, load testing, security review, and production readiness checks.

Phase 12 – Production Readiness

Validate replayability, audit completeness, data isolation, performance targets, and documentation for handoff and scaling.